

Engineering Trivia — Should Have Known That

Demo 11 of 16 · Part four, Example projects · 15–20 minutes

You need: index.html open in any modern browser, phone or projector. No network, no install, no accounts. Nothing is fetched at runtime, so it works with the wi-fi off.

Learning goals

One per stage, in the learner's own words — these are the three Takeaway cards on screen, verbatim.

1. A confident guess you commit to is what makes a correction stick — a shrug does not.
2. On a log scale, the order of magnitude is most of the answer — and item 23's rule-of-thumb number is a single point standing in for a whole risk curve.
3. A ranged or code-defined number is still worth a confident guess — precision and truth are not the same thing.

Where this sits on the path

An AI-built classroom tool from Part four: its goal is number-sense calibration, its engine is a live-scored slider running entirely on the device, and its assumptions — which of its 40 values are single facts versus ranges, minimums, or code conventions — are one tap away in each stage's Details drawer.

One question for the session

"If you had to bet, not guess — how close is your gut, really, to the numbers you already use?"

Stage 1 · Linear estimates

Question. Which of the numbers an engineer is supposed to carry around can you actually place by feel?

Predict. Drag the slider to where you think each of 35 numbers sits, before the truth is revealed.

Activity (timed beats). 0–2 min Pick a length (5/10/15), categories, and solo or teams; play the first one yourself, wrong on purpose, to set the tone. 2–10 min Poll before every lock-in if presenting to a room: hands up for higher, hands up for lower, then press **LOCK IT IN**. Read the "because..." line aloud after each reveal.

Expected observation. The band badge (Bullseye/Close/Warm/Cold/Way Off) and the "you vs. truth" markers on the track; the observation cue names the band and the miss size.

Check-yourself answer key. "It's a specified minimum — real steel usually tests higher" is correct; "it's the average strength" is the flawed model item 10 exists to break.

Teaching limit. A linear slider only works while the plausible answers fit on one linear track — which is exactly what breaks in stage 2, where the right answer to the resistivity item would sit at step 0 of 1,000. Note also that eleven items are scored against a stated range (the reveal names it): a guess inside the range is Bullseye, and the reveal's single "truth" is the anchor, not the only acceptable answer.

Stage 2 · Log estimates

Question. When a guess can be off by a factor of ten, does the exact digit even matter?

Predict. Commit to a power of ten first, on all five log-scale items (resistivity, the speed of light, the cardiac-fibrillation current, the static-shock voltage, and AA battery energy).

Activity. 10–14 min Play all five; ask the room to say a power of ten out loud before each lock-in rather than a specific number.

Expected observation. A big absolute miss can still land Bullseye if the order of magnitude is right — the opposite of stage 1's intuition.

Check-yourself answer key. "No — 100 mA is one point on a risk curve, not a safety line" is correct; treating 40 mA as automatically safe because it is under 100 mA is the refutation item's flawed model.

Teaching limit. The game cannot show WHY the curve looks the way it does (current × duration × path) — only that a single number understates it. The Details drawer names the shape; it does not re-derive IEC 60479-1's curve.

Stage 3 · Debrief

Question. Were your log guesses or your linear guesses closer?

Predict. Guess which deck scored higher before reading the comparison.

Activity. 14–18 min Read the linear-vs-log average aloud; debrief from the "should have known that" card — ask that team or student what they thought the number was and why. The wrong mental model is more useful than the right answer.

Expected observation. An average-points comparison computed live from this session's own answers — nothing here is precomputed.

Teaching limit. Five log questions is a small sample; one lucky or unlucky guess swings that average far more than the 35-question linear average moves. Say this out loud before reading the comparison as if it were a strong result.

18–20 min **Play again** from the debrief, or ↺ → **Reset** for a clean start between class periods.

Misconceptions this bank can leave, refuted

Misconception: "100 mA is the safe/unsafe line for shock." It is a rule-of-thumb anchor on a current×duration×path curve (IEC 60479-1); sustained exposure carries meaningful fibrillation risk from well below 100 mA. Item 23's qualifier and stage 2's refutation check both name this directly — it is the single highest-stakes item in the bank.

Misconception: "A36's 250 MPa is what the steel actually measures." It is ASTM A36/A36M's specified MINIMUM; real mill-certified steel commonly tests higher. Stage 1's check item exists to break exactly this substitution of a floor for a fact.

Misconception: "7¼ in risers are the code, full stop." That figure is the IRC's residential cap; the IBC's general means-of-egress cap is 7 in, and only Exception 3 for dwelling units raises it to match. Carrying the residential number into a commercial context applies the wrong code to the wrong building type.

Misconception: "13 A is safe for any 16-gauge extension cord." NEC Table 400.5(A)(1) sets 13 A by conductor count and ambient temperature, not length; a widely used practical derating guide drops well below 13 A for a long cord run. The bank's own factoid already warns a 1,500 W heater is close to the derated limit — a long cord makes it worse, not the same.

Discussion prompts

- 62.4 lb/ft³ and 0.433 psi/ft are the same fact.** Ask the room to derive one from the other ($62.4 \div 144 \text{ in}^2/\text{ft}^2$).
- Which number is the dangerous one** — 100 mA across the chest, or the few thousand volts (2,000–4,000 V for most people) it takes to feel a static shock? It is the current, not the voltage, that kills; say plainly that 100 mA is a rule of thumb, not a safe/unsafe line.
- Why not build everything out of aluminum?** Young's modulus 200 vs. 70 GPa; density 7,850 vs. 2,700 kg/m³. Both ratios are about 2.9, so stiffness-to-weight is nearly identical — the real choice is cost, buckling, and joining.
- What if steel and concrete expanded at different rates?** Every summer would crack the building — reinforced concrete works on an approximate coincidence, not an exact one.
- A36 is named after its own answer** — the 36 is the specified minimum yield strength in ksi. How many other "memorized" numbers are printed on the spec sheet's name, and how many of those are minimums rather than typicals?
- A single number can be right AND still teach the wrong lesson.** Every one of the 40 values in this bank checks out against a real source — but 23 of them ship with a qualifier because a bare number would have implied more precision than the world actually has, and 11 of the 40 are scored against a range rather than a point for the same reason. Ask the room to name one more everyday "fact" that is really a range, a floor, or a design convention.
- One gallon of gasoline is 33.7 kWh** — roughly a full day of an average US home's electricity, and (per this build's re-sourcing) a genuinely accurate figure for real gasoline, not a lowball regulatory number.

8. **Why did the log questions go so badly (or so well)?** Because they are not recall questions — getting the size of the number right is the actual skill.

Reading a Way Off: a Way Off on a **linear** question is a fact the student does not have. A Way Off on a **log** question means they do not know what *size* the answer is — say the difference out loud; it is the most transferable idea in the game.

How it is scored

Thirty-five linear questions score on **percent error**; five log questions (stage 2) score on **decades off** — $|\log_{10}(\text{guess}/\text{true})|$.

Band	Linear	Log	Pts
■ Bullseye	within 5%	1.26×	100
■ Close	within 15%	2.1×	75
■ Warm	within 35%	5×	40
■ Cold	within 60%	16×	10
■ Way Off	past that	past that	0

Because both scales share this one point table, stage 3's linear-vs-log comparison uses average points per question rather than the raw percent/decades numbers, which are not directly comparable. The debrief also prints the control — what a randomly placed slider would average on the questions actually played, scored over all 1,001 positions — and across the whole bank that control is within a few points of equal on the two decks, so a gap between a student's two averages is knowledge, not a gentler deck.

Where an item carries an accepted range (its qualifier or reveal text names it, and the Details drawer lists it with its source), a guess inside the range scores Bullseye and a guess outside it is measured from the nearer edge. Eleven items are scored this way; the other 29 score against their single value exactly as the table above describes.

Editing the question bank

The bank now lives in `src/data.js` (moved out of `index.html` for this retrofit) — each entry carries `id`, `category`, `prompt`, `units`, `value`, `min/max`, `scale` ("linear"/"log"), `factoid`, an optional `qualifier`, an optional `band`: [`lo`, `hi`], and a `source` object (`name`, `url`, `fetch`, optional `also` list for a second citation). Edit `src/data.js`, then run `node build.js` — the build derives the `qualifier` and `band` counts from the bank and refuses to run until this guide, the `README` and `demo.json` state the same numbers. A value with no `qualifier` is presented as a single, unhedged fact — add one whenever a new question is really a range, a specified minimum, or a code/editon-dependent figure, and add a `band` only where a sourced range for the same quantity already appears in the item's own text, per the standing "research, not memory" rule: back it with an actual source before shipping it, or mark it unverified.

Presenter controls

- ⚙ **Settings** holds **Open Presenter Notes** (this guide), **Presentation mode** (projector-sized type), and **Reset** (back to stage 1's setup screen, both later stages re-locked, presentation mode left as you set it).
- Stage tabs** switch stages; stages 2 and 3 stay locked until the stage before them is played through — the tabs stay visible so the room sees the whole route.
- Details** (in each stage's intro) opens that stage's sources, formula, assumptions, and boundary statement.
- Keys:** Enter/Space lock in a guess and advance the reveal; Esc closes any open dialog.

Sources (session-fetched, not recalled)

Every one of the 40 taught values carries its own citation in the app's Details drawer (per stage); this is the standards-body / primary-source subset, re-checked for this retrofit rather than carried over from memory:

- BIPM, [3rd CGPM \(1901\) Resolution 2](#) — the defined value of standard gravity.
- NIH/NCBI StatPearls, [Electrical Injuries](#) — the current thresholds behind item 23's qualifier.
- NEC Table 400.5(A)(1) as rendered by [Distributor Wire & Cable](#) (16 AWG: 13 A with two current-carrying conductors, 10 A with three), and up.codes, [IRC/IBC stair tread and riser requirements](#) (jurisdiction-routed; the page served may be a state adoption).
- 29 CFR 1926.1053(b)(5)(i), Ladders, via [Cornell LII](#) — base offset one-quarter of the ladder's *working length*, i.e. $\arccos \frac{1}{4} = 75.5^\circ$ (osha.gov's own copy refuses automated readers).
- IEC 60479-1:2018 §3.3.2 Note 1, [sample pages](#) — the 80 mA at 10 s, 50 % fibrillation figure in item 23's reveal.
- TxDOT Roadway Design Manual, [Ch. 4.8.1 Grades](#) — the terrain×design-speed structure behind item 36's qualifier.
- CPSC Publication 5098, [Tap Water Scalds](#).
- Engineering ToolBox water tables — [heat of vaporization](#) (item 29), [specific heat](#) (item 30), [boiling point vs. altitude](#) (item 25); item 33 is derived from item 5 and NIST's [231 in³ gallon](#). (The NIST WebBook page cited earlier carries none of these four numbers.)
- Engineering ToolBox, [U.S. Standard Atmosphere](#) — items 27 and 40 (replacing an ICAO Doc 7488 mirror whose TLS certificate does not validate).

The full 40-row table (value, qualifier, source, date checked) is in `src/data.js` and in this retrofit's return report; some lower-stakes items (exact unit conversions, defined SI constants) are cited to a well-established secondary reference (Engineering ToolBox, Wikipedia quoting a standard's own language, AmesWeb material tables) where the underlying standard itself is paywalled (ASTM, AISI, ACI) — named by designation in the app's Details drawer either way.

Engineering Trivia (Should Have Known That) · Bryant Harrison · Murray State University · Offline single-file web game. Open `index.html` to play. `src/demo-guide.html` is the canonical source of this document; the shipped `teacher-guide.html`, the PDF beside it, and the in-app presenter notes are all generated from it at build time.